

Library Fine Calculator using PostgreSQL

Business Problem

Libraries often face challenges in managing book returns, especially calculating fines for late submissions. Manual calculations can lead to inconsistency, inefficiency, and disputes. There is a need for an automated system that ensures fair and accurate fine assessment based on return dates.

Business Objective

To build a reliable, reusable PostgreSQL function that:

- Automatically calculates late return fines based on due and actual return dates.
- Reduces manual intervention and human error.
- Integrates smoothly with library management systems or datasets.

Success Criteria

- A working SQL function that returns accurate fines.
- Ability to test the function with various return scenarios.
- Creation of a dynamic view that lists results for all test cases.
- Exporting test data and results to CSV successfully.

Tools & Technologies

- **Database:** PostgreSQL
- **Interface:** pgAdmin 4
- **Language:** PL/pgSQL (PostgreSQL Procedural Language)
- **Data Export:** CSV (Test Cases and Results)

Dataset

A table named `test_cases` with the following structure:

id	due_date	return_date
1	2024-06-10	2024-06-15
2	2024-06-10	2024-06-10
3	2024-06-10	2024-06-08

Steps Involved

1. Create a PostgreSQL database and table `test_cases`
2. Write a stored function `calculate_fine(due_date DATE, return_date DATE)`
 - If returned late, fine = ₹5 * days_late
 - Else, fine = ₹0
3. Insert sample test cases
4. Create a view `test_case_results` to show input + calculated fine
5. Export test cases and results to CSV

Sample Function

```
CREATE OR REPLACE FUNCTION calculate_fine(due_date DATE, return_date DATE)
RETURNS NUMERIC AS $$
DECLARE
    fine NUMERIC := 0;
BEGIN
    IF return_date > due_date THEN
        fine := (return_date - due_date) * 5;
    END IF;
    RETURN fine;
END;
$$ LANGUAGE plpgsql;
```

Output View

```
CREATE VIEW test_case_results AS
SELECT
    id, due_date, return_date,
    calculate_fine(due_date, return_date) AS fine
FROM test_cases;
```

Files Included

- test_cases.csv: Sample input test data
- result.csv: Generated results using the function
- calculate_fine.sql: Function creation script

Conclusion

This project:

- Solves a real-world issue faced by libraries
- Demonstrates practical use of SQL functions and views
- Reinforces PL/pgSQL fundamentals with hands-on application
- Serves as a beginner-friendly portfolio project in data engineering and PostgreSQL

